

WEEK-7

Session-13,14

TASK-1:

To parse Input Redirection, Open Files, Redirect stdin, Handle Missing Files, Restore Streams, Test File Inputs

Code:

```
#include <stdio.h>

#include <stdlib.h>

#include <string.h>

#include <fcntl.h>

#include <unistd.h>

#include <sys/wait.h>

#define INPUT_SIZE 256

int main(void){

    char input[INPUT_SIZE];

    char filename[INPUT_SIZE];

    printf("Enter command: ");

    fgets(input, sizeof(input), stdin);

    input[strcspn(input, "\n")] = '\0';

    printf("Enter input file: ");

    fgets(filename, sizeof(filename), stdin);

    filename[strcspn(filename, "\n")] = '\0';

    int fd = open(filename, O_RDONLY);

    if (fd == -1) {

        perror("Error opening input file");

        return EXIT_FAILURE;

    }

    int saved_stdin = dup(STDIN_FILENO);

    if (saved_stdin == -1) {
```

```
    perror("dup");
    close(fd);
    return EXIT_FAILURE;
}
if (dup2(fd, STDIN_FILENO) == -1) {
    perror("dup2");
    close(fd);
    close(saved_stdin);
    return EXIT_FAILURE;
}
close(fd);
printf("\nExecuting command with redirected input...\n");
int status = system(input);
if (status == -1) {
    perror("system");
}
if (dup2(saved_stdin, STDIN_FILENO) == -1) {
    perror("restore stdin");
    close(saved_stdin);
    return EXIT_FAILURE;
}
close(saved_stdin);
printf("Input stream restored successfully.\n");
return WEXITSTATUS(status);
}
```

```

root@LAPTOP-RJFD8EBT:~# nano input_redirect.c
root@LAPTOP-RJFD8EBT:~# gcc -Wall -Wextra -g input_redirect.c -o input_redirect
root@LAPTOP-RJFD8EBT:~# echo "This is the first line." > input.txt
root@LAPTOP-RJFD8EBT:~# echo "This is the second line." >> input.txt
root@LAPTOP-RJFD8EBT:~# echo "This is the third line." >> input.txt
root@LAPTOP-RJFD8EBT:~# cat input.txt
This is the first line.
This is the second line.
This is the third line.
root@LAPTOP-RJFD8EBT:~# cat input.txt
This is the first line.
This is the second line.
This is the third line.
root@LAPTOP-RJFD8EBT:~# ./input_redirect
Enter command: cat
Enter input file: input.txt
Executing command with redirected input...
This is the first line.
This is the second line.
This is the third line.
Input stream restored successfully.
root@LAPTOP-RJFD8EBT:~# ./input_redirect
Enter command: wc -l
Enter input file: input.txt
Executing command with redirected input...
3
Input stream restored successfully.
root@LAPTOP-RJFD8EBT:~# ./input_redirect
Enter command: cat
Enter input file: missing.txt
Error opening input file: No such file or directory
root@LAPTOP-RJFD8EBT:~#

```

TASK-2:

To Parse Output Redirection, Create Output Files, Redirect stdout, Handle Permissions, Verify Output Data, Test Redirection

CODE:

```
#include <stdio.h>
```

```
#include <stdlib.h>
```

```
#include <string.h>
```

```
#include <fcntl.h>
```

```
#include <unistd.h>
```

```
#include <sys/wait.h>
```

```
#define INPUT_SIZE 256
```

```
int main(void)
```

```
{
```

```
    char input[INPUT_SIZE];
```

```
    char command[INPUT_SIZE];
```

```
    char filename[INPUT_SIZE];
```

```
    printf("Enter command with > : ");
```

```
    fflush(stdout);
```

```
if (fgets(input, sizeof(input), stdin) == NULL)
    return EXIT_FAILURE;

input[strcspn(input, "\n")] = '\0';

/* Find output redirection symbol */
char *redirect = strchr(input, '>');

if (redirect == NULL)
{
    printf("Error: Output redirection '>' not found.\n");
    return EXIT_FAILURE;
}

/* Copy command before > */
size_t command_length = (size_t)(redirect - input);

if (command_length >= sizeof(command))
    command_length = sizeof(command) - 1;

strncpy(command, input, command_length);
command[command_length] = '\0';

/* Remove extra spaces from command */
command[strcspn(command, "\n")] = '\0';

/* Copy filename after > */
strncpy(filename, redirect + 1, sizeof(filename) - 1);
filename[sizeof(filename) - 1] = '\0';
```

```
/* Remove leading spaces */
char *file_start = filename;

while (*file_start == ' ' || *file_start == '\t')
    file_start++;

if (*file_start == '\0')
{
    printf("Error: Output filename is missing.\n");
    return EXIT_FAILURE;
}

printf("Command : %s\n", command);
printf("Output file : %s\n", file_start);

/* Open/create output file */
int fd = open(file_start,
              O_WRONLY | O_CREAT | O_TRUNC,
              0644);

if (fd == -1)
{
    perror("Error opening output file");
    return EXIT_FAILURE;
}

/* Save original stdout */
int saved_stdout = dup(STDOUT_FILENO);
```

```
if (saved_stdout == -1)
{
    perror("dup");
    close(fd);
    return EXIT_FAILURE;
}
```

```
/* Redirect stdout to output file */
if (dup2(fd, STDOUT_FILENO) == -1)
{
    perror("dup2");
    close(fd);
    close(saved_stdout);
    return EXIT_FAILURE;
}
```

```
close(fd);
```

```
/* Execute command */
int status = system(command);
```

```
/* Restore original stdout */
if (dup2(saved_stdout, STDOUT_FILENO) == -1)
{
    perror("restore stdout");
    close(saved_stdout);
    return EXIT_FAILURE;
}
```

```
close(saved_stdout);
```

```
if (status == -1)
```

```
{
```

```
    perror("system");
```

```
    return EXIT_FAILURE;
```

```
}
```

```
printf("Output redirection completed successfully.\n");
```

```
printf("Output was written to: %s\n", file_start);
```

```
if (WIFEXITED(status))
```

```
    printf("Command exit status: %d\n", WEXITSTATUS(status));
```

```
return 0;
```

```
}
```

```
root@LAPTOP-RJFDBEST:~# nano output_redirect.c
root@LAPTOP-RJFDBEST:~# gcc -Wall -Wextra -g output_redirect.c -o output_redirect
root@LAPTOP-RJFDBEST:~# echo "Hello OSSP" > normal.txt
root@LAPTOP-RJFDBEST:~# cat normal.txt
Hello OSSP
root@LAPTOP-RJFDBEST:~# ./output_redirect
Enter command with > : echo Hello OSSP > output.txt
Command : echo Hello OSSP
Output file : output.txt
Output redirection completed successfully.
Output was written to: output.txt
Command exit status: 0
root@LAPTOP-RJFDBEST:~# echo Hello OSSP > output.txt
root@LAPTOP-RJFDBEST:~# cat output.txt
Hello OSSP
root@LAPTOP-RJFDBEST:~# ls -l output.txt
-rw-r--r-- 1 root root 11 Sep 18 09:03 output.txt
root@LAPTOP-RJFDBEST:~# ./output_redirect
Enter command with > : ls -l > listing.txt
Command : ls -l
Output file : listing.txt
Output redirection completed successfully.
Output was written to: listing.txt
Command exit status: 0
root@LAPTOP-RJFDBEST:~# cat listing.txt
total 924
-rw-r--r-- 1 root root 135 Aug 12 06:53 File1.txt
-rw-r--r-- 1 root root 64 Aug 8 08:42 File1.txt
-rw-r--r-- 1 root root 135 Aug 12 06:55 File2.txt
drwxr-xr-x 11 root root 4096 Aug 23 18:49 OSSP
drwxr-xr-x 2 root root 4096 Aug 20 05:42 SKILL_Week_4
-rwxr-xr-x 1 root root 21352 Sep 18 04:07 builtin_shell
-rw-r--r-- 1 root root 3613 Sep 18 04:07 builtin_shell.c
-rw-r--r-- 1 root root 7433 Sep 18 03:52 builtin_shell.c.save
-rwxr-xr-x 1 root root 21248 Sep 18 04:08 cd_navigation
-rw-r--r-- 1 root root 3603 Sep 18 04:08 cd_navigation.c
-rwxr-xr-x 1 root root 16464 Sep 16 18:51 client
-rw-r--r-- 1 root root 2123 Sep 16 18:58 client.c
-rwxr-xr-x 1 root root 16576 Aug 17 05:59 command_executor
-rw-r--r-- 1 root root 1252 Aug 17 05:59 command_executor.c
-rwxr-xr-x 1 root root 20528 Sep 18 04:55 command_history
-rw-r--r-- 1 root root 3509 Sep 18 04:55 command_history.c
-rwxr-xr-x 1 root root 16216 Aug 29 10:07 copy
-rw-r--r-- 1 root root 727 Aug 29 09:48 copy.c
-rw-r--r-- 1 root root 727 Aug 29 10:07 copy.c
-rwxr-xr-x 1 root root 16184 Aug 29 09:52 copyfile
-rw-r--r-- 1 root root 944 Aug 29 09:51 copyfile.c
-rwxr-xr-x 1 root root 18688 Sep 17 17:49 cow_demo
-rw-r--r-- 1 root root 1493 Sep 17 17:49 cow_demo.c
-rw-r--r-- 1 root root 48 Sep 17 18:18 destination.txt
-rwxr-xr-x 1 root root 17776 Sep 17 17:41 dynamic_memory
-rw-r--r-- 1 root root 1498 Sep 17 17:41 dynamic_memory.c
-rwxr-xr-x 1 root root 21424 Sep 18 04:14 export_demo
-rw-r--r-- 1 root root 48 Aug 29 08:19 text.txt
-rw-r--r-- 1 root root 8131 Sep 17 18:23 trace.txt
-rwxr-xr-x 1 root root 19376 Sep 15 03:48 variable_expand
-rw-r--r-- 1 root root 4684 Sep 18 03:48 variable_expand.c
-rwxr-xr-x 1 root root 18152 Sep 17 19:02 waitpid_demo
-rw-r--r-- 1 root root 1688 Sep 17 19:02 waitpid_demo.c
drwxr-xr-x 2 root root 4096 Aug 8 08:16 weeks
root@LAPTOP-RJFDBEST:~# ./output_redirect
Enter command with > : date > date.txt
Command : date
Output file : date.txt
Output redirection completed successfully.
Output was written to: date.txt
Command exit status: 0
root@LAPTOP-RJFDBEST:~# cat date.txt
Fri Sep 18 09:05:12 UTC 2026
root@LAPTOP-RJFDBEST:~# ls -l date.txt
-rw-r--r-- 1 root root 29 Sep 18 09:05 date.txt
root@LAPTOP-RJFDBEST:~# ./output_redirect
Enter command with > : whoami > user.txt
Command : whoami
Output file : user.txt
Output redirection completed successfully.
Output was written to: user.txt
Command exit status: 0
root@LAPTOP-RJFDBEST:~# cat user.txt
root
root@LAPTOP-RJFDBEST:~# cat user.txt
root
root@LAPTOP-RJFDBEST:~# echo Hello
Hello
root@LAPTOP-RJFDBEST:~# strace -f -e trace=openat,dup,dup2,close,read,write,execve,wait4 ./output_redirect
execve("./output_redirect", ["/output_redirect"], 0x7ffc24aa3238 /* 27 vars */) = 0
openat(AT_FDCWD, "/etc/ld.so.cache", O_RDONLY|O_CLOEXEC) = 3
close(3) = 0
openat(AT_FDCWD, "/lib/x86_64-linux-gnu/libc.so.6", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\3\0\0\0\0\0\0\0\3\0-\0\1\0\0\0\220\243\2\0\0\0\0\0....", 832) = 832
close(3) = 0
write(1, "Enter command with > : ", 23)Enter command with > : ) = 23
read(0, "\C\x038153a946b0, 1024") = 7 ERESTARTSYS (To be restarted if SA_RESTART is set)
strace: Process 7267 detached
root@LAPTOP-RJFDBEST:~# strace -f -e trace=openat,dup,dup2,close,read,write,execve,wait4 ./output_redirect
execve("./output_redirect", ["/output_redirect"], 0x7ffc24aa3238 /* 27 vars */) = 0
openat(AT_FDCWD, "/etc/ld.so.cache", O_RDONLY|O_CLOEXEC) = 3
close(3) = 0
openat(AT_FDCWD, "/lib/x86_64-linux-gnu/libc.so.6", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\3\0\0\0\0\0\0\0\3\0-\0\1\0\0\0\220\243\2\0\0\0\0\0....", 832) = 832
close(3) = 0
write(1, "Enter command with > : ", 23)Enter command with > : ) = 23
read(0, "\C\x038153a946b0, 1024") = 7 ERESTARTSYS (To be restarted if SA_RESTART is set)
strace: Process 7271 detached
root@LAPTOP-RJFDBEST:~#
```

TASK-3: To Parse Append Operator, Open Files in Append Mode, Redirect Streams, Preserve Existing Data, Validate Appends, Test Scenarios

CODE:

```
#include <stdio.h>
```

```
#include <stdlib.h>
```

```
#include <string.h>
```

```
#include <fcntl.h>
```

```
#include <unistd.h>
```

```
#include <sys/wait.h>
```

```
#define INPUT_SIZE 256
```

```
int main(void)
```

```
{
```

```
    char input[INPUT_SIZE];
```

```
    char command[INPUT_SIZE];
```

```
    char filename[INPUT_SIZE];
```

```
    printf("Enter command with >> : ");
```

```
    fflush(stdout);
```

```
    if (fgets(input, sizeof(input), stdin) == NULL)
```

```
        return EXIT_FAILURE;
```

```
    input[strcspn(input, "\n")] = '\0';
```

```
    /* Find append operator >> */
```

```
    char *redirect = strstr(input, ">>");
```

```
    if (redirect == NULL)
```

```
{
```

```

    printf("Error: Append operator '>>' not found.\n");
    return EXIT_FAILURE;
}

/* Copy command before >> */
size_t command_length = (size_t)(redirect - input);

if (command_length >= sizeof(command))
    command_length = sizeof(command) - 1;

strncpy(command, input, command_length);
command[command_length] = '\0';

/* Remove trailing spaces from command */
int len = (int)strlen(command);

while (len > 0 &&
       (command[len - 1] == ' ' || command[len - 1] == '\t'))
{
    command[len - 1] = '\0';
    len--;
}

/* Copy filename after >> */
strncpy(filename, redirect + 2, sizeof(filename) - 1);
filename[sizeof(filename) - 1] = '\0';

/* Remove leading spaces */
char *file_start = filename;

```

```

while (*file_start == ' ' || *file_start == '\t')
    file_start++;

if (*file_start == '\0')
{
    printf("Error: Output filename is missing.\n");
    return EXIT_FAILURE;
}

printf("Command   : %s\n", command);
printf("Append file: %s\n", file_start);

/*
O_APPEND:
Existing data is preserved.
New data is always written at the end.
*/
int fd = open(file_start,
              O_WRONLY | O_CREAT | O_APPEND,
              0644);

if (fd == -1)
{
    perror("Error opening append file");
    return EXIT_FAILURE;
}

/* Save original stdout */

```

```
int saved_stdout = dup(STDOUT_FILENO);
```

```
if (saved_stdout == -1)
```

```
{
```

```
    perror("dup");
```

```
    close(fd);
```

```
    return EXIT_FAILURE;
```

```
}
```

```
/* Redirect stdout to file */
```

```
if (dup2(fd, STDOUT_FILENO) == -1)
```

```
{
```

```
    perror("dup2");
```

```
    close(fd);
```

```
    close(saved_stdout);
```

```
    return EXIT_FAILURE;
```

```
}
```

```
close(fd);
```

```
/* Execute command */
```

```
int status = system(command);
```

```
/* Restore stdout */
```

```
if (dup2(saved_stdout, STDOUT_FILENO) == -1)
```

```
{
```

```
    perror("restore stdout");
```

```
    close(saved_stdout);
```

```
    return EXIT_FAILURE;
```

```
}
```

```
close(saved_stdout);
```

```
if (status == -1)
```

```
{
```

```
    perror("system");
```

```
    return EXIT_FAILURE;
```

```
}
```

```
printf("Append redirection completed successfully.\n");
```

```
printf("Data appended to: %s\n", file_start);
```

```
if (WIFEXITED(status))
```

```
    printf("Command exit status: %d\n", WEXITSTATUS(status));
```

```
return 0;
```

```
}
```

```
root@LAPTOP-RJFDBEST:~# nano append_redirect.c
root@LAPTOP-RJFDBEST:~# ^C
root@LAPTOP-RJFDBEST:~# gcc -Wall -Wextra -g append_redirect.c -o append_redirect
root@LAPTOP-RJFDBEST:~# echo "Original data" > append.txt
root@LAPTOP-RJFDBEST:~# cat append.txt
Original data
root@LAPTOP-RJFDBEST:~# ./append_redirect
Enter command with >> : echo "First appended line" >> append.txt
Command : echo "First appended line"
Append file: append.txt
Append redirection completed successfully.
Data appended to: append.txt
Command exit status: 0
root@LAPTOP-RJFDBEST:~# ^C
root@LAPTOP-RJFDBEST:~# cat append.txt
Original data
First appended line
root@LAPTOP-RJFDBEST:~# ./append_redirect
Enter command with >> : echo "Second appended line" >> append.txt
Command : echo "Second appended line"
Append file: append.txt
Append redirection completed successfully.
Data appended to: append.txt
Command exit status: 0
root@LAPTOP-RJFDBEST:~# cat append.txt
Original data
First appended line
Second appended line
root@LAPTOP-RJFDBEST:~# ./append_redirect
Enter command with >> : date >> append.txt
Command : date
Append file: append.txt
Append redirection completed successfully.
Data appended to: append.txt
Command exit status: 0
root@LAPTOP-RJFDBEST:~# cat append.txt
Original data
First appended line
Second appended line
Fri Sep 18 09:14:01 UTC 2026
root@LAPTOP-RJFDBEST:~# ./append_redirect
Enter command with >> : whoami >> append.txt
Command : whoami
Append file: append.txt
Append redirection completed successfully.
Data appended to: append.txt
Command exit status: 0
root@LAPTOP-RJFDBEST:~# cat append.txt
Original data
First appended line
Second appended line
Fri Sep 18 09:14:01 UTC 2026
root
root@LAPTOP-RJFDBEST:~# wc -l append.txt
4
root@LAPTOP-RJFDBEST:~# ./append_redirect
Enter command with >> : echo "Another line" >> append.txt
Command : echo "Another line"
Append file: append.txt
Append redirection completed successfully.
Data appended to: append.txt
Command exit status: 0
root@LAPTOP-RJFDBEST:~# wc -l append.txt
6
root@LAPTOP-RJFDBEST:~# ./append_redirect
Enter command with >> : echo Hello
Error: Append operator '>>' not found.
root@LAPTOP-RJFDBEST:~# ./append_redirect
Enter command with >> : echo Hello >>
Error: Output filename is missing.
root@LAPTOP-RJFDBEST:~# strace -f -e trace=openat,dup2,close,write ./append_redirect
openat(AT_FDCWD, "/etc/ld.so.cache", O_RDONLY|O_CLOEXEC) = 3
close(3) = 0
openat(AT_FDCWD, "/lib/x86_64-linux-gnu/libc.so.6", O_RDONLY|O_CLOEXEC) = 3
close(3) = 0
write(1, "Enter command with >> : ", 24Enter command with >> : ) = 24
echo TraceTest >> append.txt
write(1, "Command : echo TraceTest\n", 28Command : echo TraceTest
) = 28
write(1, "Append file: append.txt\n", 24Append file: append.txt
) = 24
openat(AT_FDCWD, "append.txt", O_WRONLY|O_CREAT|O_APPEND, 0644) = 3
dup2(3, 1) = 1
close(3) = 0
strace: Process 7333 attached
[pid 7333] openat(AT_FDCWD, "/etc/ld.so.cache", O_RDONLY|O_CLOEXEC) = 3
[pid 7333] close(3) = 0
[pid 7333] openat(AT_FDCWD, "/lib/x86_64-linux-gnu/libc.so.6", O_RDONLY|O_CLOEXEC) = 3
[pid 7333] close(3) = 0
[pid 7333] write(1, "TraceTest\n", 10) = 10
[pid 7333] +++ exited with 0 +++
--- SIGCHLD {si_signo=SIGCHLD, si_code=CLD_EXITED, si_pid=7333, si_uid=0, si_status=0, si_
dup2(4, 1) = 1
close(4) = 0
write(1, "Append redirection completed suc...., 43Append redirection completed successfu
) = 43
write(1, "Data appended to: append.txt\n", 29Data appended to: append.txt
) = 29
write(1, "Command exit status: 0\n", 23Command exit status: 0
) = 23
+++ exited with 0 +++
root@LAPTOP-RJFDBEST:~# cat append.txt
Original data
First appended line
Second appended line
Fri Sep 18 09:14:01 UTC 2026
root
Another line
TraceTest
root@LAPTOP-RJFDBEST:~# |
```

TASK-4:

TO parse stderr Redirection, Redirect Error Streams, Capture Error Output, Verify File Contents, Handle Failures, Test Error Cases

CODE:

```
#include <stdio.h>
```

```
#include <stdlib.h>
```

```
#include <string.h>
```

```
#include <fcntl.h>
```

```
#include <unistd.h>
```

```
#include <sys/wait.h>
```

```
#define INPUT_SIZE 256
```

```
int main(void)
```

```
{
```

```
    char input[INPUT_SIZE];
```

```
    char command[INPUT_SIZE];
```

```
    char filename[INPUT_SIZE];
```

```
    printf("Enter command with 2> : ");
```

```
    fflush(stdout);
```

```
    if (fgets(input, sizeof(input), stdin) == NULL)
```

```
        return EXIT_FAILURE;
```

```
    input[strcspn(input, "\n")] = '\0';
```

```
    /* Find stderr redirection operator */
```

```
    char *redirect = strstr(input, "2>");
```

```
    if (redirect == NULL)
```

```

{
    printf("Error: '2>' not found.\n");
    return EXIT_FAILURE;
}

/* Get command before 2> */
size_t command_length = (size_t)(redirect - input);

if (command_length >= sizeof(command))
    command_length = sizeof(command) - 1;

strncpy(command, input, command_length);
command[command_length] = '\0';

/* Remove trailing spaces */
int len = (int)strlen(command);

while (len > 0 &&
        (command[len - 1] == ' ' || command[len - 1] == '\t'))
{
    command[len - 1] = '\0';
    len--;
}

S    strncpy(filename, redirect + 2, sizeof(filename) - 1);
filename[sizeof(filename) - 1] = '\0';
char *file_start = filename;
while (*file_start == ' ' || *file_start == '\t')
    file_start++;
if (*file_start == '\0')

```

```

{
    printf("Error: Error output filename is missing.\n");
    return EXIT_FAILURE;
}

printf("Command   : %s\n", command);
printf("Error file : %s\n", file_start);
int fd = open(file_start,
               O_WRONLY | O_CREAT | O_TRUNC,
               0644);
if (fd == -1) {
    perror("Error opening file");
    return EXIT_FAILURE;
}

int saved_stderr = dup(STDERR_FILENO);
if (saved_stderr == -1){
    perror("dup");
    close(fd);
    return EXIT_FAILURE;
}

if (dup2(fd, STDERR_FILENO) == -1) {
    perror("dup2");
    close(fd);
    close(saved_stderr);
    return EXIT_FAILURE;
}

close(fd);

int status = system(command);
if (dup2(saved_stderr, STDERR_FILENO) == -1) {
    perror("restore stderr");
}

```

```

        close(saved_stderr);
        return EXIT_FAILURE;
    }

    close(saved_stderr);
    if (status == -1)
    {
        perror("system");
        return EXIT_FAILURE;
    }

    printf("stderr redirection completed successfully.\n");
    printf("Error output saved to: %s\n", file_start);
    if (WIFEXITED(status))
        printf("Command exit status: %d\n", WEXITSTATUS(status));

    return 0;
}

```

```

root@LAPTOP-RJFD8EBT:~# nano stderr_redirect.c
root@LAPTOP-RJFD8EBT:~# gcc -Wall -Wextra -g stderr_redirect.c -o stderr_redirect
root@LAPTOP-RJFD8EBT:~# ./stderr_redirect
Enter command with 2> : ls /nonexistent 2> error.txt
Command      : ls /nonexistent
Error file   : error.txt
stderr redirection completed successfully.
Error output saved to: error.txt
Command exit status: 2
root@LAPTOP-RJFD8EBT:~# cat error.txt
ls: cannot access '/nonexistent': No such file or directory
root@LAPTOP-RJFD8EBT:~# ./stderr_redirect
Enter command with 2> : cat missing.txt 2> error2.txt
Command      : cat missing.txt
Error file   : error2.txt
stderr redirection completed successfully.
Error output saved to: error2.txt
Command exit status: 1
root@LAPTOP-RJFD8EBT:~# cat error2.txt
cat: missing.txt: No such file or directory
root@LAPTOP-RJFD8EBT:~# ./stderr_redirect
Enter command with 2> : ls /nonexistent
Error: '2>' not found.
root@LAPTOP-RJFD8EBT:~# Error: '2>' not found.
Error:: command not found
root@LAPTOP-RJFD8EBT:~# ls -l stderr_redirect.c stderr_redirect error.txt error2.txt
-rw-r--r-- 1 root root  60 Sep 18 09:23 error.txt
-rw-r--r-- 1 root root  44 Sep 18 09:23 error2.txt
-rwxr-xr-x 1 root root 20128 Sep 18 09:22 stderr_redirect
-rw-r--r-- 1 root root 2788 Sep 18 09:22 stderr_redirect.c
root@LAPTOP-RJFD8EBT:~# |

```